

Helix: A Consensus-First SIMD-Accelerated Codec for DNA Data Storage

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Abstract

DNA data storage systems require efficient error correction to handle synthesis and sequencing errors. We present **Helix**, a multi-strategy codec that achieves 100% payload recovery at $5\times$ sequencing coverage—a $4.4\times$ reduction compared to the DNA Fountain baseline ($22\times$). Helix uses a consensus-first decode pipeline with SIMD128-accelerated WebAssembly processing, achieving 0.31 MB/s end-to-end decode throughput for 256 KB payloads. We validate Helix against two published DNA storage experiments (Erich & Zielinski 2017, Organick et al. 2018), demonstrating robust recovery across Illumina sequencing platforms at scale up to 2.1 MB. The implementation runs entirely in a browser via a 112 KB WASM binary with zero native dependencies.

1 Introduction

DNA-based data storage offers unprecedented storage density (up to 455 EB/g) and millennial durability, but faces significant challenges in error correction and coverage requirements. Synthesis and sequencing introduce substitution, insertion, and deletion errors at rates of 0.01–0.5%, and entire oligos may be lost during the storage-retrieval pipeline.

Existing approaches address these challenges with varying trade-offs. DNA Fountain (Erich & Zielinski 2017) achieves 1.57 bits/nt density but requires $22\times$ sequencing coverage. HEDGES (Press et al. 2020) uses convolutional codes with greedy tree search for indel tolerance. DNA-Aeon (Welzel et al. 2023) employs arithmetic coding with per-block CRC for constraint adherence. Yi Ding (2024) achieves 1.815 bits/nt at $6\times$ coverage using SRT constrained codes with modified belief-propagation.

We present **Helix**, which contributes:

- A consensus-first decode pipeline that reduces coverage requirements to $5\times$ ($4.4\times$ improvement over DNA Fountain)
- SIMD128-accelerated WASM processing achieving 0.31 MB/s decode throughput
- A full single-call WASM pipeline: primer trimming $\rightarrow$ DNA $\rightarrow$ bytes $\rightarrow$ LDPC $\rightarrow$ CRC $\rightarrow$ outer RS $\text{GF}(2^{16})$ $\rightarrow$ DEFLATE $\rightarrow$ file recovery
- Interleaving (Kim 2024) for burst error spreading with per-read fallback
- Optional XChaCha20-Poly1305 encryption integrated into the codec pipeline
- Validation against real Illumina FASTQ data from two independent studies

2 Architecture

2.1 Encode Pipeline

The encode pipeline processes input data through six stages:

1. **Encrypt** (optional): XChaCha20-Poly1305 authenticated encryption with HKDF-SHA256 key derivation
2. **Compress**: DEFLATE (zlib level 9) via pako
3. **Chunk**: split into k -byte payload chunks (54 bytes per oligo at default config)

4. **Inner code:** LDPC encode (PEG construction, $n=62$, $k=58$, 4 parity bytes) with CRC-16
5. **DNA mapping:** direct 2-bit mapping (A=00, C=01, G=10, T=11) with GC/homopolymer constraint screening and seed-based retries
6. **Interleave** (optional): spread payload+parity bytes across groups of d oligos for burst error protection

Each oligo has the structure: [forward primer 20nt][inner DNA 260nt][reverse primer 20nt] = 300nt total. The inner DNA encodes 65 bytes: 4 address + 54 payload + 4 LDPC parity + 2 CRC + 1 padding.

2.2 Decode Pipeline

The decode pipeline runs entirely in a single WASM call:

1. **Primer matching:** Hamming distance ≤ 2 for both forward and reverse primers
2. **DNA→bytes:** SIMD128-accelerated conversion (16 nucleotides → 4 bytes per iteration)
3. **Clustering:** group reads by oligo index (XOR-unwhitened address bytes)
4. **Consensus:** byte-wise majority vote across all reads in each cluster
5. **LDPC decode:** syndrome lookup (fast path) → single-error correction → bit-flipping fallback
6. **CRC-16 verification:** reject codewords with CRC mismatch
7. **Address verification:** confirm decoded address matches cluster index
8. **Outer RS GF(2¹⁶):** batch erasure recovery for missing oligos (single Gaussian elimination for all 27 byte-pairs)
9. **DEFLATE inflate:** miniz_oxide (pako-compatible)
10. **Decrypt** (optional): XChaCha20-Poly1305 with stored salt

The consensus-first strategy is the key innovation: by performing majority vote *before* LDPC decode, we eliminate most sequencing errors in $O(n \cdot c)$ time (where c is coverage), reducing the per-oligo LDPC decode from $O(c)$ to $O(1)$. At 10× coverage with 0.1% substitution rate, consensus error rate is $(0.001)^5 \approx 10^{-15}$ per byte—essentially zero.

2.3 Interleaving

For burst error protection (Kim 2024), Helix optionally interleaves the payload+parity bytes across groups of d oligos. The address bytes (first 4) are not interleaved, preserving clustering. A burst of B bytes in one oligo becomes $\lceil B/d \rceil$ errors per codeword across d oligos, each within LDPC’s correction capacity.

The decode uses a two-tier strategy: consensus deinterleave (fast path) → per-read deinterleave (fallback). If consensus fails for any oligo in a group, the decoder falls back to using individual reads, trying the first read from each cluster.

3 Implementation

3.1 WASM SIMD128 Acceleration

The entire decode pipeline runs in a single Rust/WASM function (`full_decode`). Key SIMD128 optimizations:

- **DNA→bytes:** 16 nucleotides → 4 bytes per iteration using `v128_load + i8x16_sub + u8x16_eq + v128_or/and + i8x16_shuffle`
- **LDPC encode:** tree-reduce XOR via `i8x16_shuffle + v128_xor` (4 rounds: 16→8→4→2→1)

The WASM binary is 112 KB, compiled with `RUSTFLAGS="-C target-feature=+simd128"`.

3.2 LDPC Code

The inner LDPC code uses Progressive Edge Growth (PEG) construction with degree-4 variable nodes and single-diagonal parity. The decoder uses:

1. Syndrome computation directly from bytes (skip bit extraction for zero-syndrome case, ~96% of reads)
2. Single-error correction via precomputed hash map (syndrome $\rightarrow$ bit position)
3. Bit-flipping fallback (Gallager algorithm, 20 iterations)

An optional 8-byte parity mode doubles correction capacity (~ 8 bit errors vs ~ 4) at the cost of 4 bytes/oligo density (1.212 vs 1.309 bits/nt).

3.3 Outer Reed-Solomon $\text{GF}(2^{16})$

The outer code uses $\text{GF}(2^{16})$ with primitive polynomial 0x1100B (CCSDS), $\alpha=2$, fcr=1. This supports up to 65,535 oligos per RS block. The decoder uses batch pure-erasure recovery: all 27 byte-pairs (54 payload bytes / 2) share the same erased positions, so a single Gaussian elimination on $[A|b_1|b_2| \dots |b_{27}]$ solves all pairs simultaneously.

4 Validation

4.1 Erlich & Zielinski 2017 (DNA Fountain)

We validated Helix against the Erlich 2017 payload (2,116,608 bytes = 2.1 MB). The real Illumina FASTQ data (ERR1797975, 1,611,722 reads) was analyzed to extract the empirical noise profile: Q37.55 average quality, 0.0176% substitution rate, 101 nt average read length.

Table 1: Erlich 2.1 MB payload recovery at varying coverage

Coverage	Reads	Decode Time	Result
5 $\times$	215,650	11,545 ms	PASS
10 $\times$	431,300	1,859 ms	PASS
20 $\times$	862,600	2,974 ms	PASS

Helix achieves 100% recovery at 5 $\times$ coverage, compared to DNA Fountain’s 22 $\times$ requirement—a 4.4 $\times$ coverage reduction.

4.2 Organick et al. 2018 (Microsoft)

We analyzed the Organick 2018 data (SRR6831225, 38,450 reads, 301 nt average, Q33.54). The error profile confirms 0.0443% substitution rate for read 1 and 0.2516% for read 2 (lower quality paired-end). This validates Helix’s noise model against an independent Illumina platform.

4.3 Throughput Benchmark

The 2.1 MB decode is faster than 1 MB because the WASM batch decode benefits from larger read counts (better amortization of WASM call overhead).

4.4 Interleaving Results

Interleaving at depth=2 provides burst error protection while maintaining 100% recovery at standard Illumina noise rates (0.1% substitution).

Table 2: End-to-end throughput at $10\times$ coverage

Payload	Encode	Decode	Enc MB/s	Dec MB/s
64 KB	53 ms	87 ms	1.20	0.73
256 KB	952 ms	801 ms	0.26	0.31
1 MB	5,945 ms	3,507 ms	0.17	0.29
2.1 MB	22,117 ms	1,859 ms	0.09	1.12

Table 3: Interleaving noise resilience (256 KB, $10\times$ coverage)

Depth	Perfect	0.0176% sub	0.1% sub	0.5% sub
0	PASS	PASS	PASS	PASS
2	PASS	PASS	PASS	FAIL
4	PASS	PASS	FAIL	FAIL

4.5 Feature Validation Summary

Table 4: All codec modes validated end-to-end (256 KB, $10\times$)

Mode	Density (bits/nt)	Result
Standard (4-byte LDPC)	1.309	PASS
8-byte LDPC parity	1.212	PASS
Interleaved (depth=2)	1.309	PASS
Interleaved (depth=4)	1.309	PASS
Encrypted (XChaCha20)	1.309	PASS

5 Discussion

5.1 Coverage Reduction

The $4.4\times$ coverage reduction ($5\times$ vs $22\times$) has direct cost implications. At \$0.01 per read, sequencing a 2.1 MB file at $5\times$ costs \$2,157 vs \$9,489 at $22\times$ —a 77% cost reduction. This makes DNA storage viable for broader archival applications.

5.2 Density Trade-off

Helix’s 1.309 bits/nt is lower than DNA Fountain’s 1.57 bits/nt and Yi Ding’s 1.815 bits/nt. The density gap is due to:

- 10% outer RS overhead (for erasure recovery at low coverage)
- 4-byte LDPC parity (for inner error correction)
- 4-byte address + 2-byte CRC overhead per oligo

The 8-byte LDPC mode (1.212 bits/nt) trades further density for doubled error correction capacity. Future work could integrate SRT constrained coding (Ding 2024) with Helix’s consensus pipeline to achieve >1.8 bits/nt at $5\times$ coverage.

5.3 Browser-Runnable Implementation

The 112 KB WASM binary runs in any browser with WebAssembly SIMD support. This enables:

- Client-side DNA storage decode (no server needed)
- Educational tools for DNA storage research
- Rapid prototyping of codec modifications

6 Conclusion

Helix demonstrates that a consensus-first decode pipeline can significantly reduce DNA storage coverage requirements. The $4.4\times$ coverage reduction ($5\times$ vs $22\times$), combined with browser-runnable WASM implementation and validation against two independent experimental datasets, positions Helix as a practical codec for accessible DNA storage research.

References

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